

Osmosis & The Shrinking Potato

Investigating how water moves across plant cells using potato strips, salt solutions, rulers & balances!

Kabir's Kitchen Story: Kabir noticed that cucumber slices became droopy and soft when placed in salted water! Why does salt water change the texture and weight of fresh vegetables? Let's conduct a quantitative laboratory experiment to find out!

Essential Materials Required

 **Potato**

 **3 Solutions**
(Pure Water, Light Salt,
Concentrated Salt)

 **Ruler**
(Length Measurement in
mm)

 **Balance**
(Mass Measurement in
grams)

 **3 Beakers / Jars**
& Paper Towels

1. Detailed Experimental Procedure & Variables

Setup Variable	Solution A (Pure Water)	Solution B (Light Salt Solution)	Solution C (Concentrated Salt Solution)
Liquid Solution	100% Pure Tap / Distilled Water	Water + 0.5 tbsp Salt	Water + 2 tbsp Salt
Potato Core Dimensions	Identical ~50 mm length	Identical ~50 mm length	Identical ~50 mm length
Initial Measurements	Measure mass with balance (g) & length with ruler (mm)	Measure mass with balance (g) & length with ruler (mm)	Measure mass with balance (g) & length with ruler (mm)
Soaking Time	30–45 Minutes	30–45 Minutes	30–45 Minutes
Final Measurements	Blot dry, re-weigh on balance & remeasure length with ruler	Blot dry, re-weigh on balance & remeasure length with ruler	Blot dry, re-weigh on balance & remeasure length with ruler
Texture Observation	Stiff, hard, turgid (firm)	Slightly flexible / neutral	Soft, limp, droopy, flaccid

2. Scientific Principle: What is Osmosis?

Osmosis is the movement of water molecules through a semi-permeable membrane from an area of higher water concentration (lower solute concentration) to an area of lower water concentration (higher solute concentration).

- **In Pure Water (Solution A):** Water moves *into* the potato cells, increasing cell turgor pressure, length, and total mass.
- **In Concentrated Salt Solution (Solution C):** Water moves *out* of the potato cells into the salt water to balance concentrations, causing the cells to shrink, lose mass, and become limp.

3. Osmosis in Everyday Life

 **Raw Mango Pickle (Achar)**
Salt draws water out of mangoes via osmosis, preventing spoilage and shrinking the fruit.

 **Soaking Plump Raisins**
Dry raisins swell up when placed in water as water diffuses inside through their skin.

STUDENT PRACTICE & EXPERIMENTAL ASSESSMENT SHEET

Student Name: _____ Date: _____

Section A: Quantitative Data Collection & Step-by-Step Procedure

Step-by-Step Laboratory Protocol:

- Step 1: Cut 3 uniform potato strips using a cutter. Record initial texture.
- Step 2: Use a **ruler** to measure initial length (\$L_1\$ in mm). Use a **balance** to measure initial mass (\$M_1\$ in grams).
- Step 3: Submerge one potato strip into each of the **3 solutions** (Pure Water, Light Salt, Concentrated Salt) for 30 minutes.
- Step 4: Remove strips, blot gently with paper towels, and remeasure length (\$L_2\$) with ruler and mass (\$M_2\$) with balance.
- Step 5: Calculate change in mass (\$\Delta M = M_2 - M_1\$) and percentage change.

Q1. Data Record Table (Fill in your laboratory observations below):

Solution Type	Initial Length (mm) (Ruler)	Final Length (mm) (Ruler)	Initial Mass (g) (Balance)	Final Mass (g) (Balance)	Change in Mass (g) (Final - Initial)	Final Texture (Firm / Limp)
Solution A (Pure Water)						
Solution B (Light Salt)						
Solution C (Conc. Salt)						

Section B: Multiple Choice & Concept Checks

Q2. Which tool is used to record the mass change of the potato strips before and after soaking?

- ☐ A) Measuring tape
- ☐ B) Digital Balance
- ☐ C) Thermometer
- ☐ D) Graduated cylinder

Q3. Why does a potato strip decrease in mass and length when placed in a concentrated salt solution?

- ☐ A) Salt directly dissolves the potato starch.
- ☐ B) Water moves out of the potato cells into the salt solution via osmosis.
- ☐ C) Salt enters the potato and forces air out.
- ☐ D) Water enters the potato cells and causes them to burst.

Q4. During osmosis, water molecules naturally migrate from:

- ☐ A) An area of higher water concentration to lower water concentration.
- ☐ B) An area of lower water concentration to higher water concentration.
- ☐ C) A dry region to an even drier region.
- ☐ D) The potato skin to the core only.

Section C: Short Answer & Experimental Analysis

Q5. Explain how using a ruler and a balance provides clearer scientific evidence of osmosis than just feeling the potato texture with your hands.

Q6. Predict what would happen to the mass reading on the balance if a limp potato strip from Solution C is placed back into Solution A (Pure Water) for 45 minutes. Explain why.